

UNIT 5: Introduction to Strength of Materials

Syllabus topics: Normal & shear stresses; stress-strain diagrams (ductile & brittle materials); elastic constants; Hooke's law; relation $E = 2G(1+\nu)$; one-dimensional loading of members of varying cross-section; principal stresses & maximum shear stress.

Question 1

5a.	State the Hooke's law and draw the typical stress-strain curve for a ductile and brittle material.	3	CO5	K1, K2
5b	A steel rod of 22mm diameter is subjected to a tension force of 75 kN. Knowing that Poisson's ratio is 0.3 and the modulus of elasticity is 200 GPa, determine (a) the elongation of the rod along the 200-mm length, (b) the change in diameter of the rod.	4	CO5	K1, K2, K3, K4
5c	For the given state of stress $\sigma_x = 50$ MPa, $\sigma_y = 10$ MPa and $\tau_{xy} = 15$ MPa, determine (a) the principal normal stresses, (b) the maximum shear stress.	3	CO5	K1, K2, K3, K4, K5

NME 101 | 2025-26 | Q5(a) Hooke's law & stress-strain curve, Q5(b) Elongation & change in diameter of steel rod, Q5(c) Principal stresses & max shear stress

Question 2

	Figure 4	Figure 5			
5(a)	Derive the expression $E=2G(1+\nu)$ relating modulus of elasticity(E) with modulus of rigidity (G). Poisson's ratio is denoted as ν .		5	CO5	K6
5(b)	At a critical point in a machine member is subjected to tensile stress of 500 N/m^2 in x-direction, compressive stress of 1000 N/m^2 in y-direction and shear stress 250 N/m^2 in xy plane. Find the principal plane, principal stresses and maximum shear stress.		5	CO5	K4, K5

END SEM | NME-102 | 2022-23 | Q5(a) Derive $E = 2G(1+\nu)$, Q5(b) Principal plane, principal stresses & max shear stress

Question 3

	Figure 4.	Figure 5.			
5	(a) A member ABCD is subjected to point loads P_1 , P_2 , P_3 and P_4 as shown in Fig. 6. Calculate the force P_2 necessary for equilibrium, if $P_1 = 45 \text{ kN}$, $P_3 = 450 \text{ kN}$ and $P_4 = 130 \text{ kN}$. Determine the total elongation of the member, assuming the modulus of elasticity to be $2.1 \times 10^5 \text{ N/mm}^2$.		5+5	CO5	K4, K2
	Figure 6.				
	(b) State Hooke's law. Define the terms: Elasticity, elastic limit, Young's modulus and modulus of rigidity.				

End Semester | Q5(a) Member ABCD – force P_2 & total elongation, Q5(b) Hooke's law, elasticity, Young's modulus, modulus of rigidity